AWS EC2 Windows Server Lab

Step-by-step guide: Launch an EC2 instance and connect using Remote Desktop Protocol (RDP)

Project objective

Create a Windows Server virtual machine on Amazon EC2 through the AWS Management Console, configure secure RDP access, retrieve the initial Windows administrator password, and connect from a Windows computer.

Services and tools

Amazon EC2 · Windows Server 2025 Base AMI · EC2 key pair · Security group · Remote Desktop Connection (mstsc)

Evidence included

Screenshots from the actual lab are included throughout this guide. They document the console, instance configuration, security group, running instance, password retrieval screen, and successful RDP session.

Prepared as a portfolio project guide

1. Lab overview

Amazon EC2 provides resizable virtual servers in AWS. In this lab, a Windows Server instance is launched in the Asia Pacific (Mumbai) region (ap-south-1). Remote Desktop Protocol uses TCP port 3389 to connect to the graphical Windows desktop.

Observed configuration in the supplied screenshots

Item	Lab configuration
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Instance name	windows-rdp-lab
Operating system	Microsoft Windows Server 2025 Base
Instance type	t3.micro
Key pair	windows-rdp-key
Security group	windows-rdp-sg
Inbound rule	RDP / TCP / 3389 / My IP (122.171.21.98/32)
Network	Default VPC and selected subnet; auto-assign public IP enabled
Outcome	Instance running; 3/3 status checks passed; RDP desktop opened

Note: The public IP shown in the source screenshots is intentionally not reproduced here. Public IP addresses and account details should be redacted before posting portfolio evidence.

2. Step-by-step implementation

Step 1 — Open the EC2 dashboard

Sign in to the AWS Management Console, search for EC2, and open the EC2 dashboard. Confirm the selected region is Asia Pacific (Mumbai).

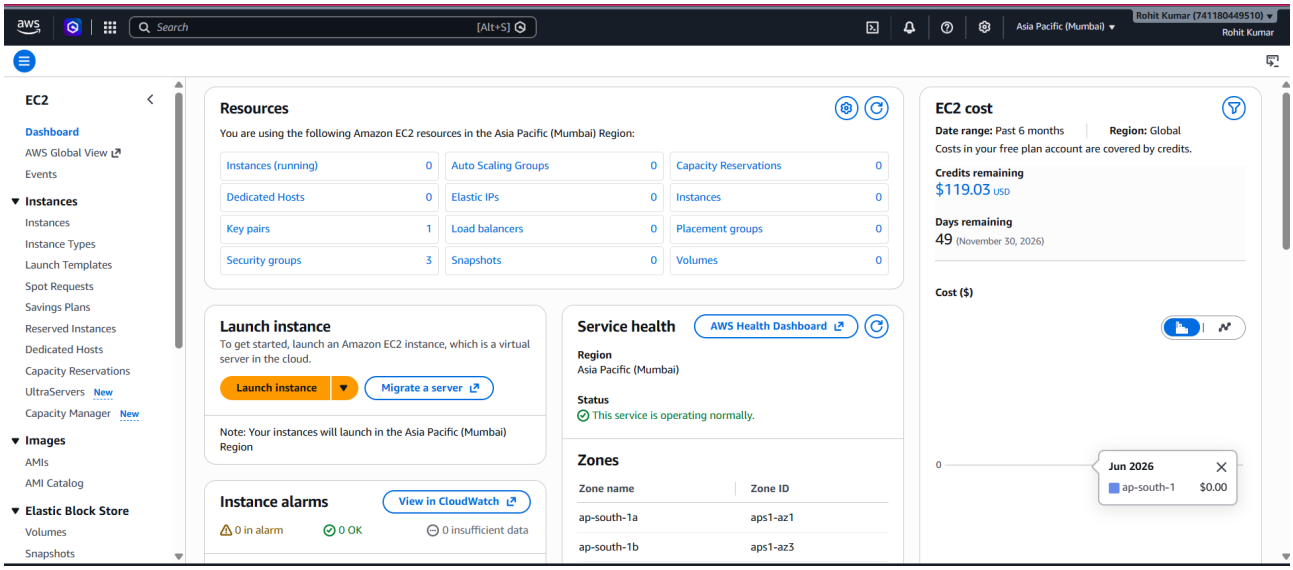


Figure 1. EC2 dashboard showing the Mumbai region and the Launch instance entry point.

Step 2 — Configure the Windows EC2 instance

Set the instance name to **windows-rdp-lab**. Select the Windows Server 2025 Base AMI and the t3.micro instance type as shown in the lab. Select the existing key pair **windows-rdp-key** or create a new key pair if starting from scratch. Download the private key and keep it outside the repository.

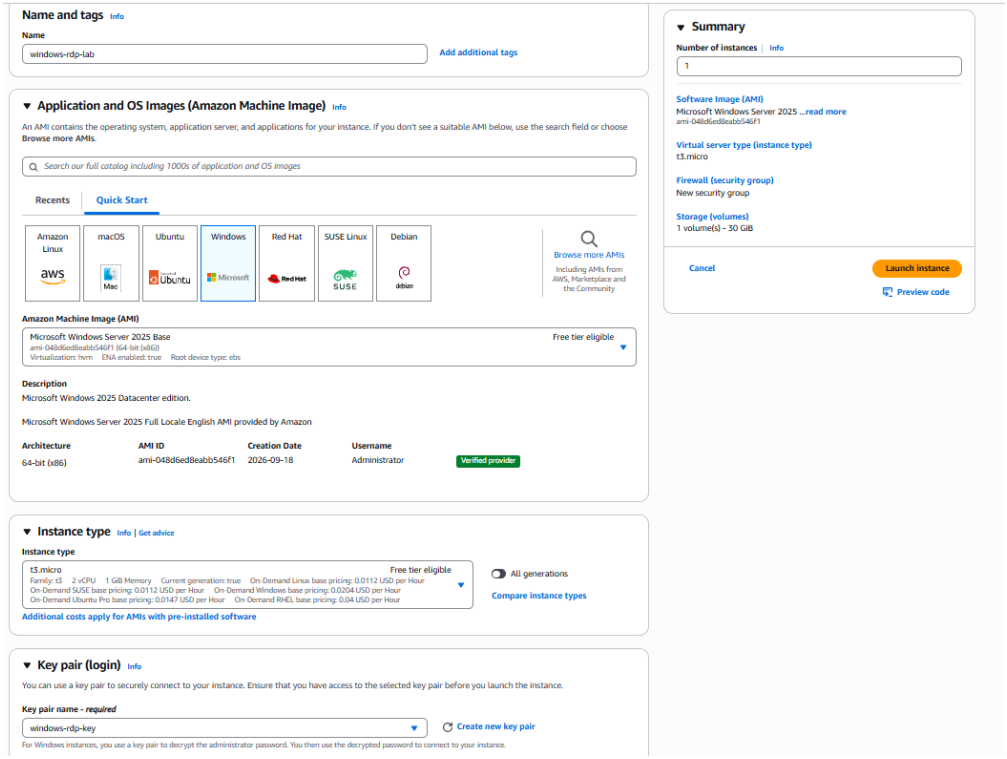


Figure 2. Instance name, Windows Server image, instance type, and key-pair selection.

Step 3 — Configure networking and the security group

Use a VPC and subnet with a route to the internet, enable public IPv4 assignment for this direct-RDP lab, and create a security group named **windows-rdp-sg**. Add an inbound RDP rule on TCP port 3389 with source set to **My IP**. In the supplied screenshot, the rule is restricted to a single /32 IP address. Do not use 0.0.0.0/0 for public RDP access.

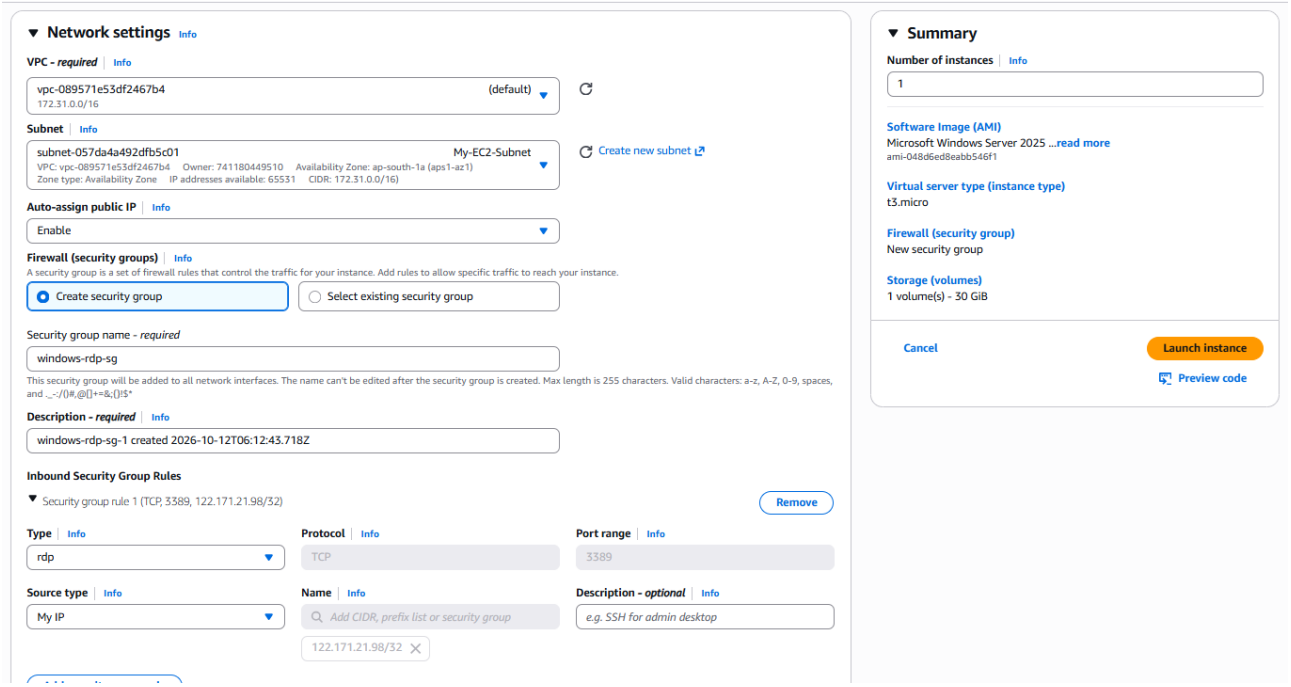


Figure 3. Network settings and RDP rule restricted to My IP.

Step 4 — Launch and verify the instance

Click Launch instance. In EC2 → Instances, wait until the instance state is Running and status checks pass. Record the instance ID privately if needed for troubleshooting; avoid publishing account-specific identifiers unnecessarily.

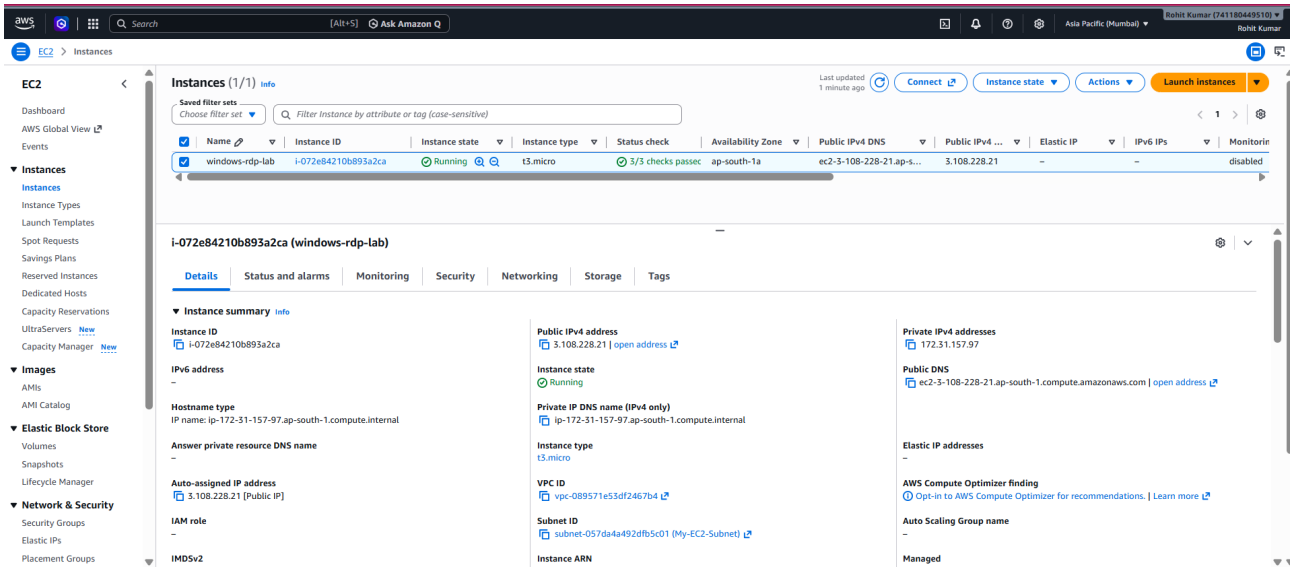


Figure 4. The instance is running and status checks have passed.

Step 5 — Retrieve the Windows password

Select the instance and choose Connect → RDP client → Get password. Upload the private key file (.pem) or provide its contents as prompted, then decrypt the initial Windows administrator password. The username is commonly **Administrator**. The password is a secret: do not include it in screenshots, GitHub files, or LinkedIn posts.

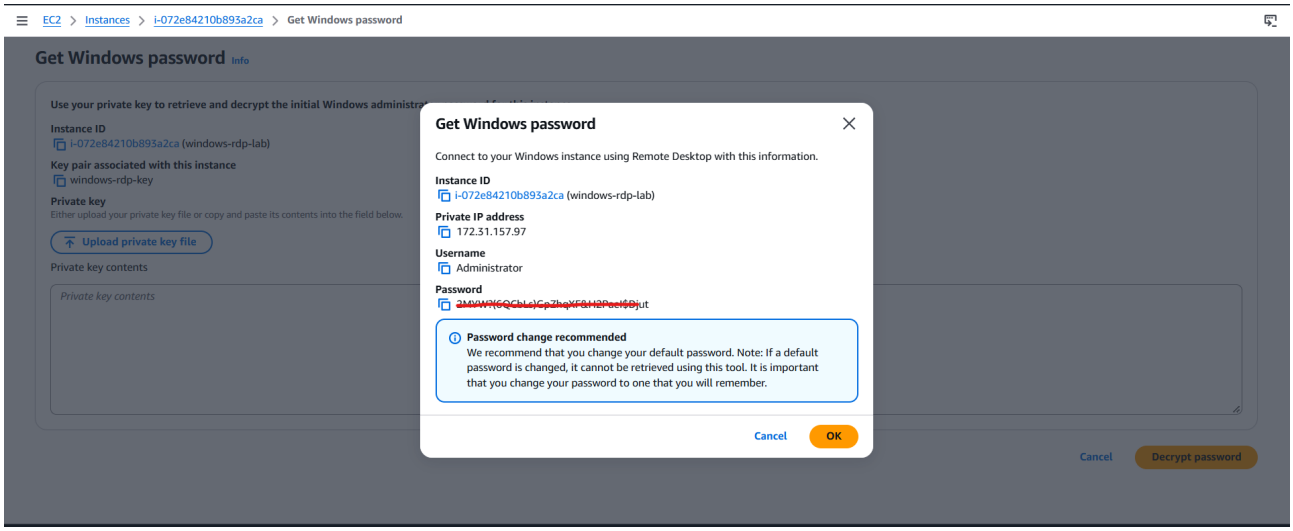


Figure 5. Windows password retrieval screen. The supplied screenshot contains a visible password; this figure is deliberately omitted from the public-facing guide. Keep the original image private and never upload it.

Step 6 — Connect using Remote Desktop

On a Windows computer, press Win + R, type **mstsc**, and press Enter. Enter the instance public IPv4 address, click Connect, and sign in as Administrator using the decrypted password. Verify that the Windows desktop opens successfully.

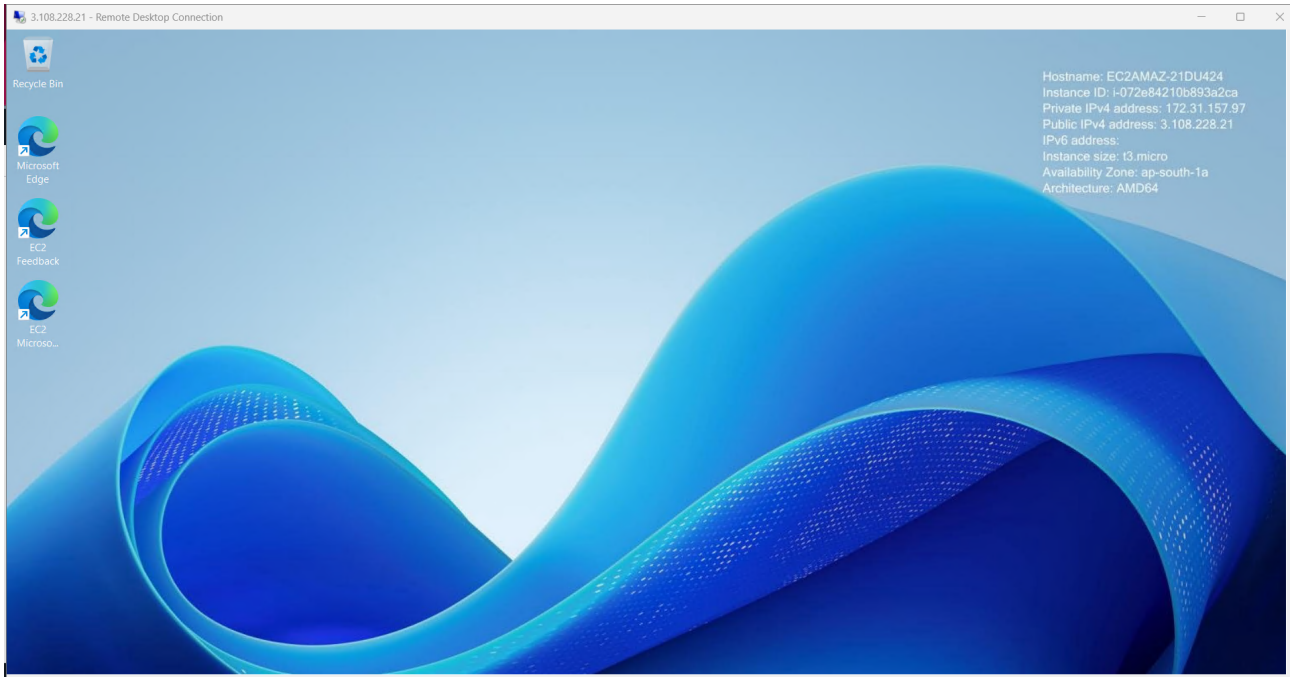


Figure 6. Successful Remote Desktop session to the Windows EC2 instance.

3. Security best practices

• Restrict TCP 3389 to your current public IP rather than opening it to everyone.
• Keep the .pem private key out of GitHub, cloud-shared folders, screenshots, and public posts.
• Never publish the decrypted Administrator password, even briefly.
• Redact public IP addresses, account IDs, and other sensitive console details from public evidence.
• For longer-term administration, consider AWS Systems Manager or another secure access design instead of exposing RDP directly.
• Terminate the instance when the lab is complete if you no longer need it, and review associated resources for ongoing charges.

4. Troubleshooting

Problem	Checks
Password is not available	Wait for Windows initialization to finish, then retry Get password with the correct private key.
RDP connection times out	Verify the instance is running, has a public IPv4 address, and the security group allows TCP 3389 from your cur
Credentials rejected	Use the username Administrator and the decrypted password for this exact instance. Check keyboard layout an
Your IP changed	Update the security group source to your current public IP using My IP.
Unexpected cost	Check EC2 instance, EBS volume, Elastic IP and other resource charges. Terminate unneeded resources.